

Data and Artificial Intelligence

Cyber Shujaa Program

Week 5 Assignment

Data Visualization Using Tableau

(HR Analytics Dashboard)

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1. Introduction

This assignment focused on developing practical skills in data visualization using Tableau through the analysis of Human Resource (HR) data. The goal was to create an interactive HR Analytics Dashboard that helps the HR manager make informed decisions regarding employee performance, demographics, and salary distribution.

The dashboard consolidates various HR metrics, enabling data-driven insights into workforce structure and trends. Through this project, I enhanced my understanding of Tableau's capabilities in data connection, cleaning, calculation, and visualization.

2. Objectives

The main objectives of the assignment were to:

- Gain hands-on experience using Tableau for data visualization and analytics.
- Understand HR business requirements and identify key performance indicators (KPIs) such as total employees, average salary, and tenure.
- Clean and transform HR data for visualization.
- Create calculated fields for Age Group, Average Salary, and Total Employees.
- Design interactive dashboards for management-level insights.
- Publish the dashboard on Tableau Public for professional presentation.

3. Tasks Completed

Step 1: Understanding Business Needs

The HR department required a visual system to monitor:

- Workforce demographics (gender, education level, department composition).
 - Salary distribution and performance ratings.
 - Hiring trends and employee retention.
- The dashboard was divided into key sections for Overview, Demographics, and Income Analysis.

Step 2: Data Preparation and Cleaning

The dataset from wk05-Tableau-HR-Project-Materials.zip was loaded into Tableau. Data preparation involved:

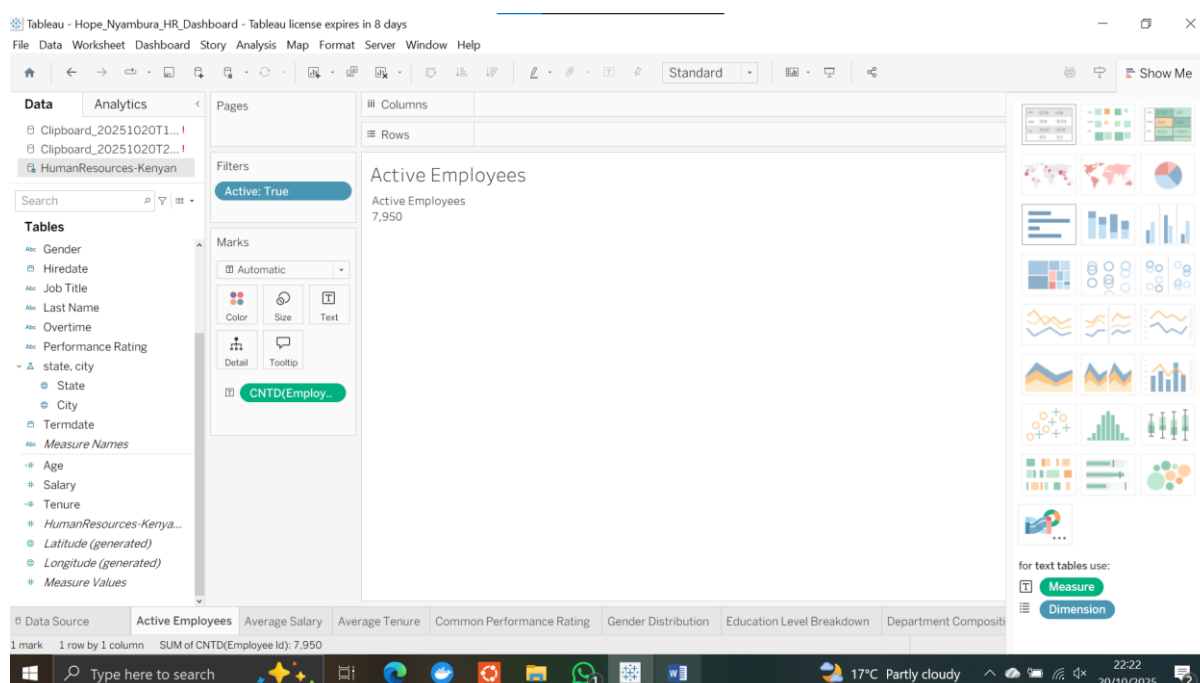
- Removing null and duplicate records.
- Converting date fields into valid formats.
- Splitting employee names into first and last names.
- Ensuring numeric fields like *Salary* and *Age* were correctly typed.
- Creating relationships between the Employee and Department tables.

Step 3: Creation of Calculated Fields

To enhance analysis, the following calculated fields were created:

Measure Name	Formula / Description
Age Group	IF [Age] < 30 THEN "Under 30" ELSEIF [Age] < 40 THEN "30–39" ELSEIF [Age] < 50 THEN "40–49" ELSE "50+" END
Total Employees	COUNTD([Employee ID])
Average Salary	AVG([Salary])
Department Salary Gap	AVG([Salary]) - WINDOW_AVG(AVG([Salary]))
Gender Ratio	COUNTD([Employee ID]) grouped by [Gender]

These calculated fields were used in charts and KPIs across the dashboards.



Step 4: Visualizations Created

Multiple visualizations were designed to represent the HR insights clearly and professionally:

- **Bar Charts** – Gender Distribution, Salary by Education Level, Salary vs. Performance Rating.
- **Map** – Employee locations across Kenya.
- **Line Chart** – Salary Trend by Hire Date.
- **Horizontal Bars** – Average Salary by Department, Department Composition, Common Performance Rating.
- **KPIs (Cards)** – Total Employees, Active Employees, Average Salary, Average Tenure.

Each visualization was enhanced with consistent color schemes, tooltips, and formatting for clarity and readability.

Step 5: Dashboard Design and Interactivity

Overview Dashboard (Main View)

- Displays Total Employees (8,950), Active Employees (7,950), Average Salary (70,933), and Average Tenure (1.34 years).
- Includes a Salary Trend Line showing hiring and salary evolution from 2016–2025.
- Provides quick insights through charts such as:
 - Gender Distribution
 - Salary by Education Level
 - Average Salary by Department
 - Common Performance Ratings
 - Employee Location Map (Kenya)

The design uses a horizontal container layout for alignment and balanced spacing, ensuring KPIs and charts are visually coherent.

Overview Dashboard

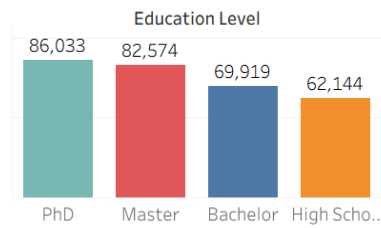
Total Employees
8,950

Active Employees
7,950

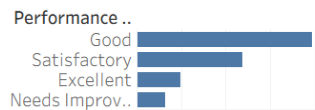
Average Salary
70,933

Average Tenure
1.339

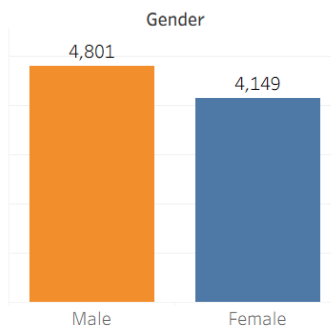
Salary by Education Level



Common Performance Rating

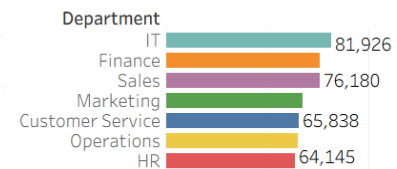


Gender Distribution

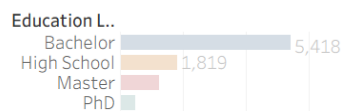


Gender
Male
Female

Average Salary by Department



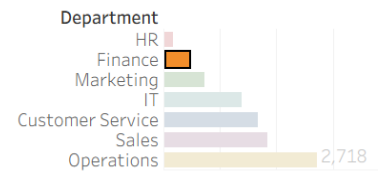
Education Level Breakdown



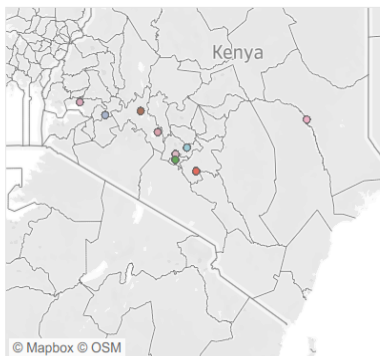
Education Level
Bachelor
High School
Master
PhD

Department
Customer Servi..
Finance
HR
IT
Marketing
Operations
Sales

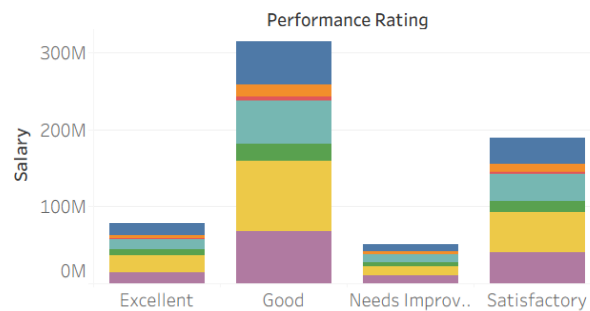
Department Composition



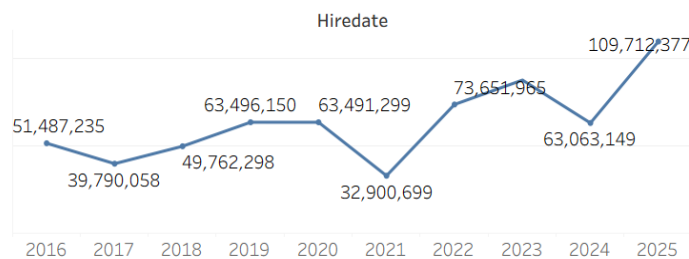
Employee Location



Salary vs Performance Rating



Salary Trend by Hire Date



Step 6: Publishing and Sharing

After completion, the dashboard was saved and shared for accessibility through Google Drive.

1. The final Tableau workbook was saved locally after verification of all visuals.
2. It was then uploaded to Google Drive for storage and sharing purposes.
3. The dashboard was titled “HR Analytics Dashboard”.
4. A shareable Google Drive link was generated to allow easy access and review.

4. Conclusion

This project provided comprehensive exposure to Tableau as a Business Intelligence tool. I learned how to:

- Connect and clean data efficiently.
- Develop calculated fields to derive meaningful KPIs.
- Create interactive and visually appealing dashboards.

The final HR Analytics Dashboard successfully transformed raw data into actionable insights about workforce demographics, performance, and salary trends. This assignment significantly improved my confidence in data storytelling and using Tableau for data-driven decision-making.

5. Link to Published Tableau Dashboard

[<https://drive.google.com/file/d/1BlhAbpZCgfv0yER5tp20Y6VwbV1J2OYJ/view?usp=sharing>]